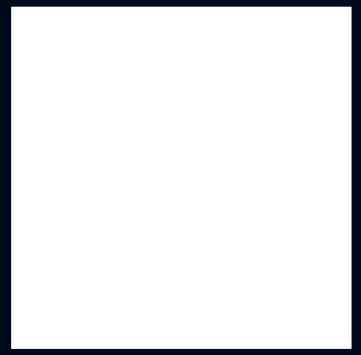


# Ambient Systems

Climate Management for Cities of Tomorrow

# Modern Buildings
# Face Many Challenges

1. Climate Change
2. Urban Pollution
3. Indoor Contaminants
4. Rising Energy Costs
5. Increasing Technological Complexity



# Indoor Farming Opportunity

1. Vertical Farming is a rapidly growing industry with 24% CAGR*.

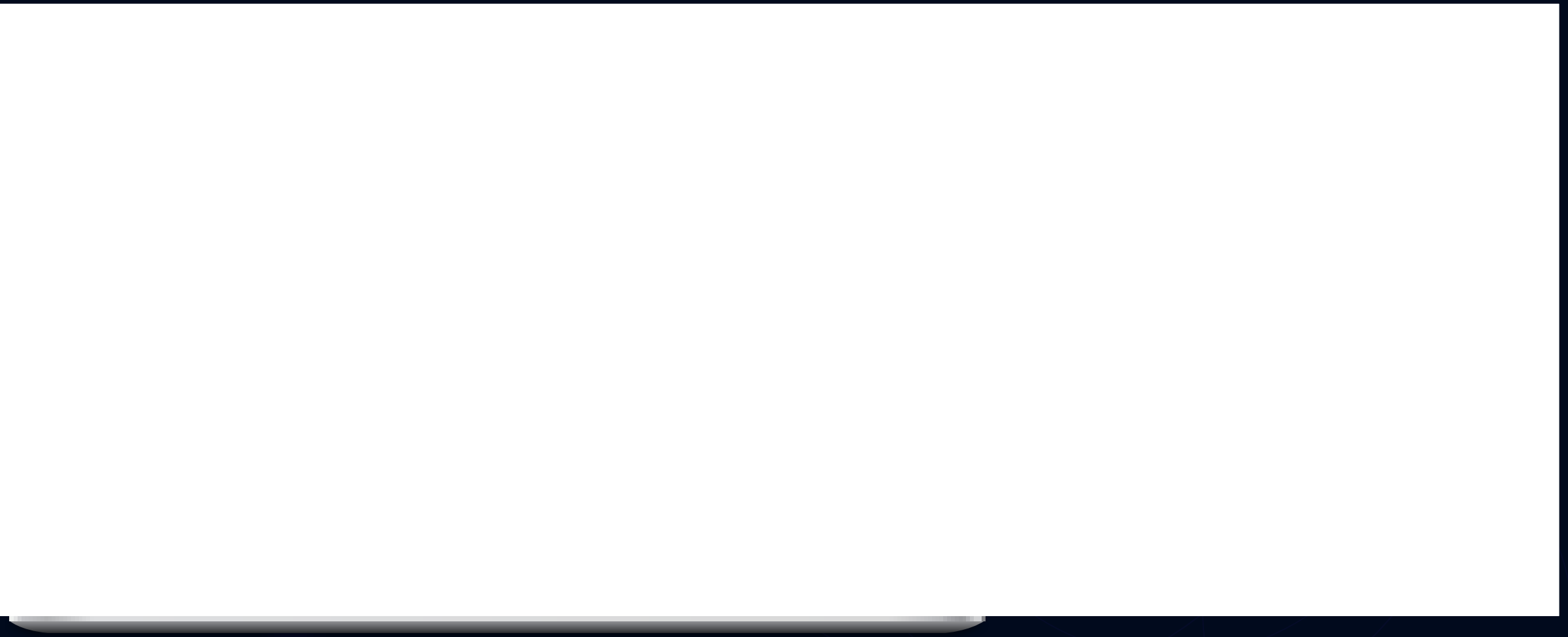
2. Asia-Pacific is a global leader in vertical farming with market value of >$350M.

3. Clear need from several farm operators in Singapore.

*Source: Statista

# The Urban Farmer's Dilemma





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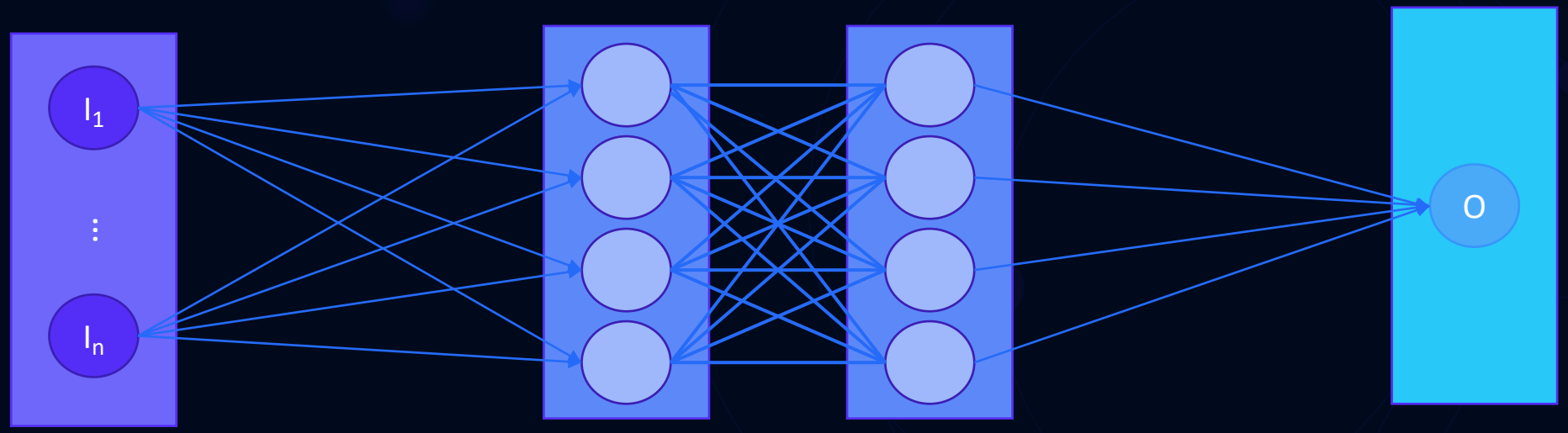


# Ambient Systems

Climate Management for Cities of Tomorrow

# Automation Engine

## Neural Network based on Building Science




The sensors and external data are used as inputs. They are passed down with each subsequent stop or layer extracting a different set of high-level features.

layers determine more complex feature, such as the optimal equipment setpoints to achieve desired outputs.

### Output Layer

Farm operator will receive recommendations in real-time on what the optimal equipment set points should be for the building equipment.

# Intellectual Property

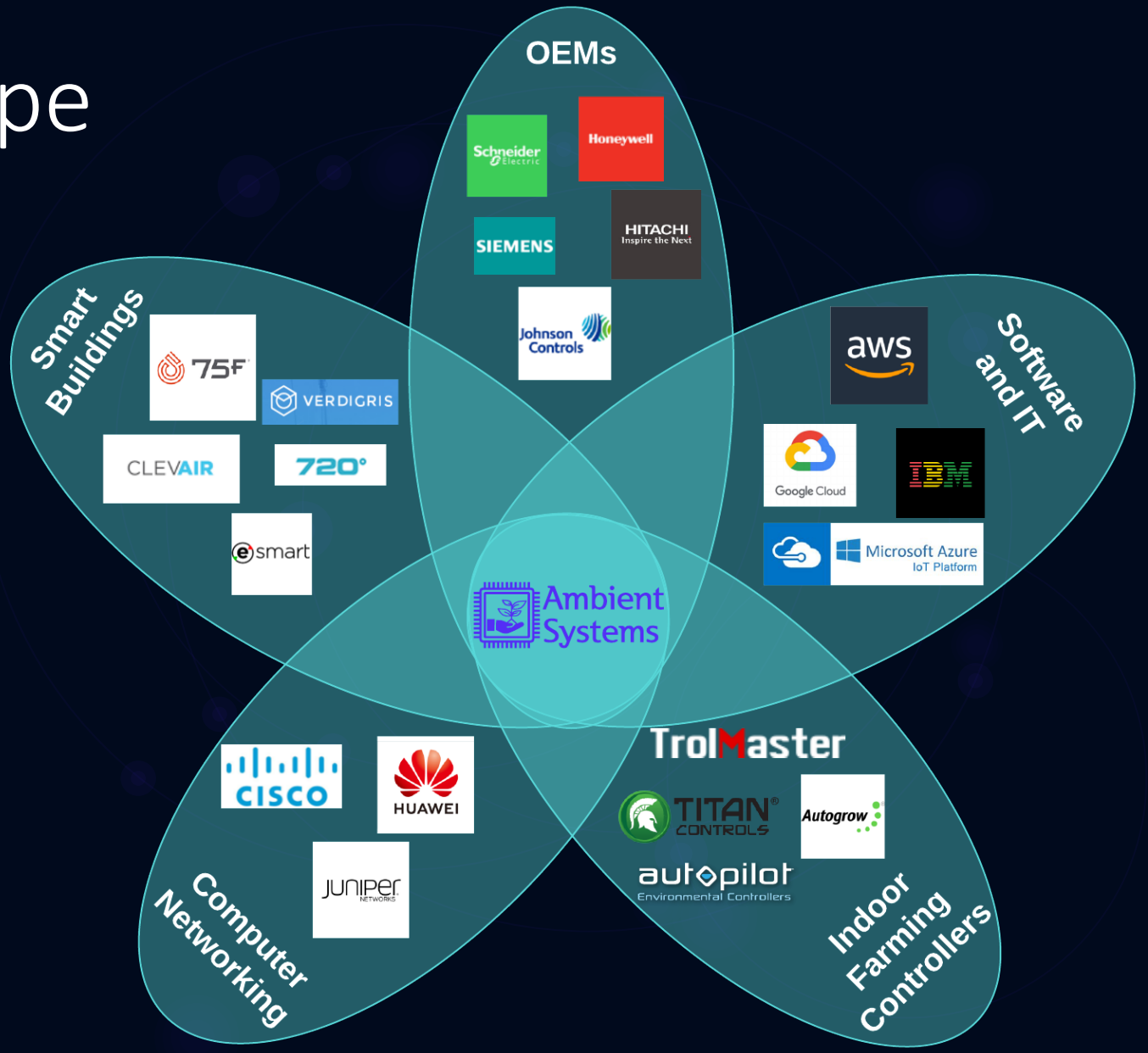
1. Patent: Automated Climate Management for Indoor Farming Facilities

2. Copyright: Proprietary Database Design and Algorithms for the Automation Engine

3. Trade Secret: Operational Data on Creating an Optimal Indoor Climate

# Competitive Landscape

Ambient is positioned at the intersection of multiple segments in order to provide a tailored solution for the indoor farming beachhead market.



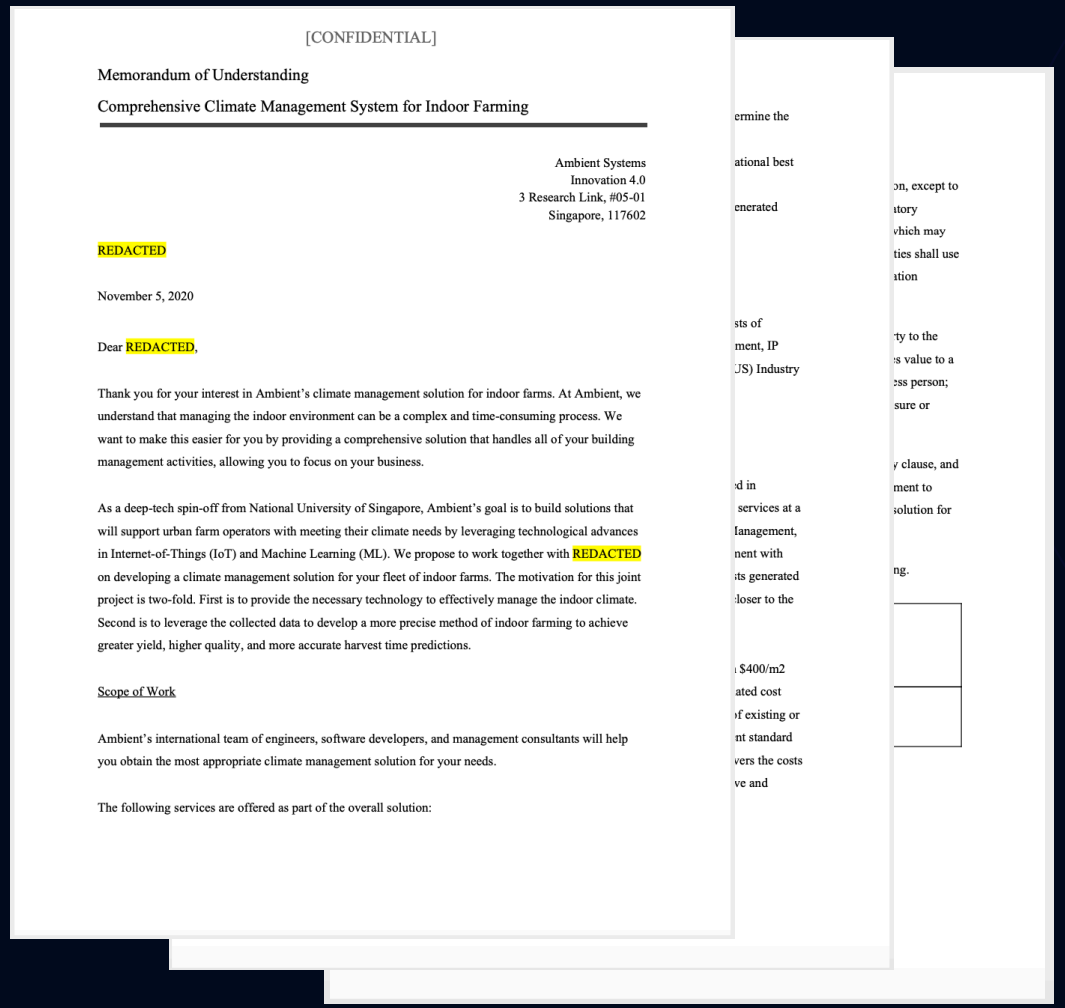
# Competitive Benchmark

| | Ambient | Autogrow | 75F | Johnson Controls |
|---|---|---|---|---|
| **Segment** | Multiple | Indoor Farming Controller | Smart Buildings | OEM |
| **Proximity to Customer (Low Logistics Cost)** | ✓ Yes | ✗ No | ✗ No | ✓ Yes |
| **Indoor Farming Features** | ✓ Yes | ✓ Yes | ✗ No | ✗ No |
| **Modern Web-based UI** | ✓ Yes | ✗ No | ✓ Yes | ✗ Some |
| **Wireless IoT Network** | ✓ Yes | ✗ No | ✓ Yes | ✓ Yes |
| **Advanced Optimization Methods** | ✓ Machine Learning | ✗ No | ✗ Some | ✗ Some |
| **Pricing** | Quotation (based on area) | $3000 for one unit + Installation Fees | Quotation | Quotation |

# Innovative Revenue Model

| Section | Component | Cost |
|---|---|---|
| **CAPEX** | | |
| **Services** | Audit and Design | Complimentary |
| | Installation | $4,000 - $5,000 |
| **Hardware** | Ambient Smart Sensors | $1,500 - $2,500 |
| | Ambient Edge Nodes | $2,500 - $4,500 |
| | Ambient Control Unit | $4,000 - $6,000 |
| **CAPEX SubTotal** | | **$12,000 - $18,000** |
| **OPEX** | | |
| **Operations** | Secure Data Management | Complimentary |
| | System Maintenance | Complimentary |
| | Parts Replacement | Complimentary |
| **Ambient Platform** | Facility Analytics | Value-sharing |
| | ACMV Optimization | Value-sharing |
| | Predictive Maintenance | Value-sharing |
| | Yield Prediction | Value-sharing |
| | Marketplace | Value-sharing |
| **OPEX SubTotal** | | **$0** |
| **Estimated Total Cost** | | **$12,000 - $18,000 + OPEX value-sharing** |

# Customer Traction



[CONFIDENTIAL]

Memorandum of Understanding

Comprehensive Climate Management System for Indoor Farming

Ambient Systems
Innovation 4.0
3 Research Link, #05-01
Singapore, 117602

REDACTED

November 5, 2020

Dear REDACTED,

Thank you for your interest in Ambient's climate management solution for indoor farms. At Ambient, we understand that managing the indoor environment can be a complex and time-consuming process. We want to make this easier for you by providing a comprehensive solution that handles all of your building management activities, allowing you to focus on your business.

As a deep-tech spin-off from National University of Singapore, Ambient's goal is to build solutions that will support urban farm operators with meeting their climate needs by leveraging technological advances in Internet-of-Things (IoT) and Machine Learning (ML). We propose to work together with REDACTED on developing a climate management solution for your fleet of indoor farms. The motivation for this joint project is two-fold. First is to provide the necessary technology to effectively manage the indoor climate. Second is to leverage the collected data to develop a more precise method of indoor farming to achieve greater yield, higher quality, and more accurate harvest time predictions.

Scope of Work

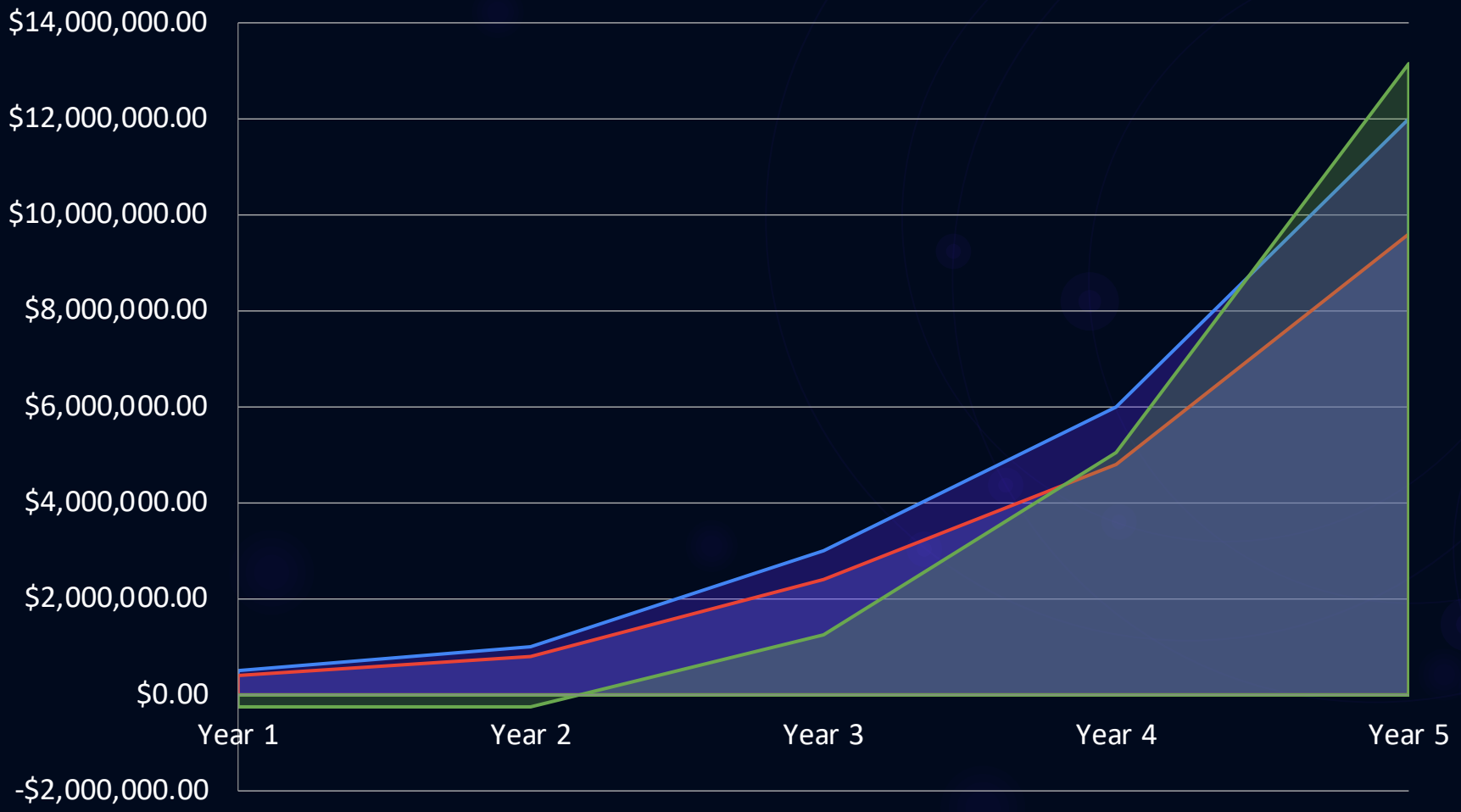
Ambient's international team of engineers, software developers, and management consultants will help you obtain the most appropriate climate management solution for your needs.

The following services are offered as part of the overall solution:

**Summary**:

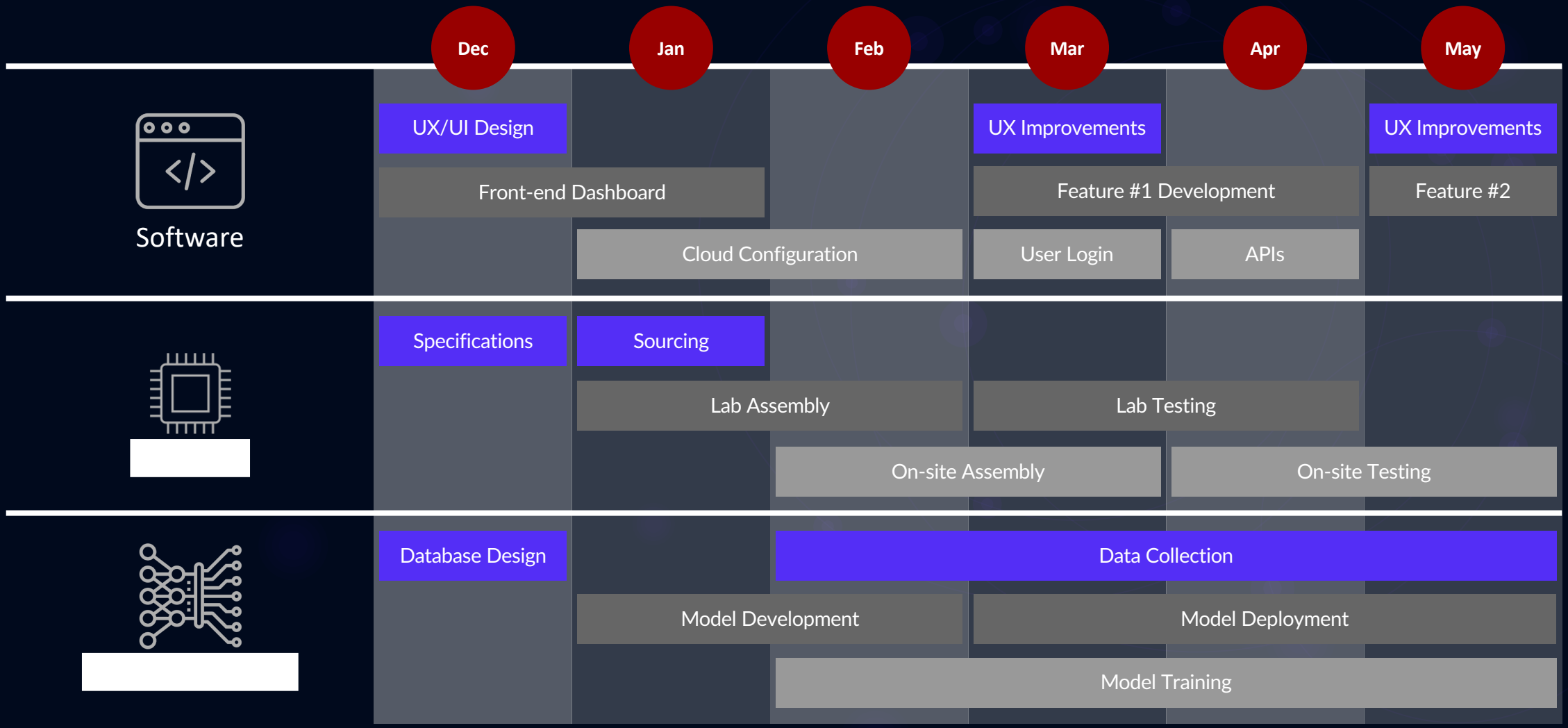
1. Signed an MoU with farm operator in Singapore
2. In conversation with two leading farm operators
3. Identified gap for similar solution in the healthcare sector

# Financials




Funding Gap: $650k
Valuation: $8.5M
Discount Rate: 10%

# MILESTONES


Dec
Jan
Feb
Mar
Apr
May
Software
UX/UI Design
Front-end Dashboard
Cloud Configuration
UX Improvements
Feature #1 Development
User Login
APIs
UX Improvements
Feature #2
Specifications
Sourcing
Lab Assembly
On-site Assembly
Lab Testing
On-site Testing
Database Design
Model Development
Data Collection
Model Deployment
Model Training

# Ambient Systems

An international team consisting of experienced entrepreneurs, engineers, and management consultants with a wealth of experience across major markets.

*Product Development*

Computer Science Master's Student
NUS (Singapore)

*Corporate Development*

Raised over $23M for ventures in highly regulated industries in Canada & EU

TBD

*Market Development*




# Summary

1. Ambient has identified a competitive product-market fit as a smart-building technology provider for indoor farms.
2. Initial customer is interested to move foreword with the pilot project.
3. Asking for $100k seed funding to expand customer base.

Contact: ivan@ambientsystems.ai